

Hospital–Insurer Integration and Hospital Quality

Ira Abraham*

Alberto Cappello†

[Click here for the latest version](#)

September 26, 2026

Abstract

We study how vertical integration between hospitals and insurers affects hospital quality. Our setting is Medicare Advantage, where hospitals increasingly sponsor their own health plans and thereby become financially exposed to the downstream utilization of beneficiaries enrolled in those plans. Exploiting within-hospital changes in plan-sponsorship status over time and quasi-random variation in hospital assignment generated by ambulance referral patterns, we find little evidence of an average effect of plan sponsorship on 30-day unplanned readmissions in the full sample. In unconcentrated hospital markets, however, admission to a plan-sponsoring hospital reduces the probability of a 30-day unplanned readmission by 9.57 percentage points. These findings suggest that, when hospital markets are more competitive, hospital–insurer integration strengthens incentives to improve hospital quality and reduce avoidable readmissions.

Keywords: Hospital Readmission, Provider-Sponsored Health Plans, Vertical Integration, Medicare Advantage.

*University of Iowa (ira-abraham@uiowa.edu)

†Yale University (alberto.cappello@yale.edu)

1 Introduction

Provider-sponsored health plans (PSHPs) are health plans sponsored by healthcare providers and represent a form of integration between healthcare financing (the plan side) and healthcare delivery (the provider side). Provider-sponsored health plans have become increasingly prevalent among U.S. hospitals. The number of hospitals owning or jointly owning a provider-sponsored health plan increased from 692 (18.3% of hospitals) in 2018 to 834 (27.2%) in 2023 (Chen et al., 2026). Proponents argue that PSHPs can better align incentives between payers and providers, facilitate care coordination, and potentially improve efficiency and quality (Frakt et al., 2013; Meyers et al., 2020). Critics emphasize that provider–insurer integration may also create incentives to steer patients toward affiliated providers, weaken competition, and increase market power, with gains from integration therefore not necessarily translating into better patient outcomes (Cuesta et al., 2024; Lin et al., 2021; Bejarano et al., 2024a).

This paper provides new empirical evidence on the effects of hospital–insurer integration on hospital quality by exploiting within-hospital changes in plan-sponsorship status over time. We study large hospital organizations that sponsor Medicare Advantage (MA) plans and ask whether becoming a plan sponsor changes hospital quality, as measured by 30-day unplanned readmissions. Because MA plans receive capitated payments from Medicare, plan sponsorship exposes the integrated hospital system to the downstream utilization of beneficiaries enrolled in its plan and can therefore strengthen incentives to reduce avoidable hospital use. A readmission is defined as an unplanned inpatient hospitalization occurring within 30 days of discharge from an earlier hospital stay. Readmissions are economically important in Medicare: in 2018, there were 16.9 readmissions per 100 Medicare index admissions, with an average cost of approximately \$15,500 per readmission (Weiss and Jiang, 2021).

We refer to hospitals that sponsor a plan as vertically integrated (VI) hospitals. VI hospitals may respond to the financial incentives in two ways. First, they may make costly upfront investments in quality and care-management capacity—for example, hiring nurses and case managers, improving discharge planning, medication reconciliation, post-discharge monitoring, and coordination with outpatient providers—that reduce the expected probability and cost of subsequent readmissions. Because many such investments involve hospital-wide protocols and organizational changes, their effects may extend beyond patients enrolled in the hospital’s own plan. Second, hospitals may respond by changing admission decisions rather than improving underlying quality. As Gupta (2021) shows in the context of the Hospital Readmissions Reduction Program, hospitals can reduce measured readmissions by changing whether patients who return to the emergency department are admitted as inpatients or

treated as outpatients. We therefore focus on emergency episodes, where patient sorting into hospitals is substantially more limited. Although federal emergency-care requirements constrain hospitals’ ability to refuse emergency patients who present for care, hospitals retain discretion over the subsequent disposition of marginal patients, including whether to admit them as inpatients, place them under observation, or discharge them after outpatient treatment. This distinction allows us to study whether hospital plan sponsorship leads primarily to improvements in care that prevent patients from returning to the hospital or instead to changes in admission behavior once patients return.

We address two conceptually distinct sources of endogeneity to study the effect of VI hospitals on readmission rates. The first source of endogeneity is supply-side selection. Hospitals that differ in the VI status may differ in other quality dimension, that affect readmission risk. To address this concern, we exploit the two-year structure of our data in a difference-in-differences framework. Specifically, we compare changes in readmission outcomes within the same hospital over time as its vertical integration status changes.

The second concern is patient-side selection: the composition of beneficiaries admitted to a hospital may change when the hospital’s VI status changes. If healthier patients are systematically more likely to choose VI hospitals, but this sorting pattern is constant over time, hospital fixed effects absorb the resulting time-invariant differences in patient composition. However, if the composition of patients changes when a hospital becomes vertically integrated, within-hospital comparisons may still be biased. Because we do not want to impose the assumption that patient sorting remains unchanged around VI adoption, we complement the fixed-effects design with an instrumental-variables strategy. Following the logic of [Doyle Jr et al. \(2015\)](#), we construct an ambulance-company-based instrument that exploits differences in ambulance referral patterns across emergency episodes to generate plausibly exogenous variation in admission to VI hospitals.

A large literature in health economics and industrial organization shows that hospital performance and quality incentives depend on market structure. Competition can strengthen hospitals’ incentives to invest in quality and operational performance, while hospitals with greater market power may face weaker pressure to undertake costly quality-improving investments ([Kessler and McClellan, 2000](#); [Gaynor et al., 2015](#); [Cooper et al., 2022](#); [Chandra et al., 2016](#)). This consideration is particularly relevant in our setting because hospital plan sponsorship creates a new financial incentive to reduce downstream utilization, but hospitals must still choose whether to respond to that incentive by undertaking costly investments in discharge planning, care coordination, and post-discharge management. In more competitive hospital markets, the financial incentive created by plan sponsorship may be reinforced by greater competitive pressure to improve quality, increasing the return to hospital-wide

investments that reduce readmissions. In more concentrated markets, by contrast, hospitals with greater market power may face weaker competitive pressure to translate the financial incentives created by integration into improvements in care delivery. Vertical integration and hospital competition may therefore be complementary: plan sponsorship increases the financial return to reducing avoidable utilization, while competition increases the incentive to act on that opportunity. This framework generates the prediction that the reduction in readmissions associated with hospital plan sponsorship may be larger in unconcentrated hospital markets.

We construct a pooled sample of two years of administrative data on Medicare Advantage beneficiaries. We observe beneficiaries’ plan enrollment and use the AHRQ’s Compendium of Health Systems to determine whether an MA plan is sponsored by a hospital organization. We use MA claims-level data to obtain information on patient location, the category of the emergency event, ambulance transportation, the admitting hospital, the principal diagnosis, inpatient length of stay, and the 30-day all-cause unplanned readmission event. We implement an IV design embedded in a difference-in-difference linear probability model. Since we only have two years of data, we do not have individual level data to check for pre-trends in our DID. As a proxy for the DID, we use a long panel dataset of hospital-level readmission rate for Medicare Fee-for-Service beneficiaries. We document that hospital that later become a plan sponsor are not following a different pre-trend compared to standard hospitals.

After accounting for patient selection, we find little evidence of an average hospital-side effect in the full sample. However, once we focus on hospitals operating in unconcentrated markets, patients admitted to a plan-sponsoring hospital have a 9.57 percentage point lower readmission probability than those admitted to standard hospitals. This pattern is robust to a stricter definition of vertically integrated hospitals based on sponsor-plan enrollment relative to bed capacity. We also find that VI hospitals have shorter inpatient length of stay, suggesting that the readmission effects reflect more efficient discharge planning, care coordination, or post-acute management rather than longer inpatient stays.

As far as we know, we are the first to provide causal evidence for how incentives of an integrated hospital affect hospital quality. Our paper contributes to four strands of literature. First, we contribute to the large literature studying how managed care affects health care spending, utilization, and patient outcomes. A central question in this literature is whether managed care can reduce health care spending without adversely affecting quality. [Agafiev Macambira et al. \(2026\)](#) find that Private managed care reduced overall healthcare costs by 5.6% and that savings were driven largely by pharmacy utilization management—reducing overuse and substituting lower-cost prescription drugs without lowering quality. [Cutler et al. \(2000\)](#) find that HMO patients with heart disease have substantially

lower health care expenditures than patients in traditional insurance, with little difference in treatments or health outcomes. In the Medicare setting, [Baicker et al. \(2013\)](#) show that greater Medicare Advantage (MA) penetration reduces hospital costs and length of stay, including through spillovers to patients outside MA. [Afendulis et al. \(2017\)](#) find that an expansion of MA reduces hospital utilization and mortality, while [Curto et al. \(2019\)](#) document substantially lower health care spending in MA relative to Traditional Medicare, driven primarily by lower utilization rather than lower prices. More broadly, this literature demonstrates that managed care and insurer incentives can meaningfully affect the quantity and organization of care. However, much less is known about what happens when the insurer and the hospital system are under common ownership. We contribute to this literature by studying whether a hospital system’s decision to sponsor an MA plan changes hospital-side care delivery and patient outcomes, as measured by 30-day unplanned readmissions.

Second, our paper contributes to the literature on how hospitals respond to financial incentives tied to patient outcomes. Much of the existing evidence comes from the Hospital Readmissions Reduction Program (HRRP), which introduced financial penalties for hospitals with excess readmissions. Several studies document substantial declines in readmission rates following the introduction of the program [Zuckerman et al. \(2016\)](#); [Desai et al. \(2016\)](#). [Gupta \(2021\)](#) provides causal evidence that hospital responses to HRRP penalties account for a substantial share of the observed decline in readmissions and shows that hospitals responded along two margins: improvements in quality and changes in the admission of patients returning to the hospital.¹ We extend this literature by studying a different source of financial incentives. Rather than an explicit penalty tied to readmission performance, hospital sponsorship of an MA plan changes the financial return to downstream utilization. Because MA plans receive capitated payments, a health system that sponsors a plan becomes financially exposed to the subsequent utilization of beneficiaries enrolled in its plan. This can increase the return to hospital investments in discharge planning, care coordination, and other activities that prevent costly readmissions. We study whether these altered incentives translate into hospital-wide changes in care delivery and readmission risk.

A related literature documents spillovers from managed-care incentives to patients outside the plans directly subject to those incentives. [Glied and Zivin \(2002\)](#) show that physician treatment patterns depend on the managed-care composition of the physician’s practice, suggesting that practice-level changes in care can spill over across insurance types. [Chernew et al. \(2008\)](#) find that greater Medicare HMO penetration reduces spending among beneficiaries who remain in traditional Medicare. Most directly related to our setting, [Baicker](#)

¹The magnitude of the causal effect of HRRP remains debated; see, for example, [Ody et al. \(2019\)](#) and [Ziedan and Kaestner \(2021\)](#).

[et al. \(2013\)](#) find that increases in Medicare Advantage penetration reduce hospital utilization and costs among patients outside Medicare Advantage, consistent with hospital-level changes in treatment patterns. Our paper differs from this literature by studying whether hospital ownership of an MA plan changes the quality of care delivered by the hospital itself, as measured by 30-day unplanned readmissions.

Finally, we contribute directly to the emerging literature on vertical integration between health systems and Medicare Advantage plans. Existing evidence is informative but largely descriptive or associational. [Park et al. \(2023\)](#) show that integrated MA plans have higher star ratings and lower non-claims costs than non-integrated plans, although they also charge higher premiums. [Bejarano et al. \(2024b\)](#) document the growth of vertically integrated MA plans and show that legacy-integrated plans tend to have higher star ratings and higher premiums than both non-integrated and newer integrated plans. On the patient-experience margin, [Bejarano et al. \(2024a\)](#) find that integrated MA plans overall are not associated with meaningfully better care experiences than non-integrated plans, with the strongest advantages concentrated among legacy-integrated plans. [Meyers et al. \(2020\)](#) study inpatient care among beneficiaries enrolled in provider-integrated MA plans and find differences in patterns of inpatient care and marginally lower mortality, but no difference in readmissions. More recent evidence from [Ma et al. \(2026\)](#) similarly documents differences in post-acute utilization, spending, and patient outcomes between hospital-operated and other MA plans, but relies on cross-sectional comparisons. Taken together, this literature suggests that hospital-insurer integration may affect costs, quality, and patterns of care, but does not establish whether becoming a plan sponsor causally changes the quality of care delivered by the hospital itself. We fill this gap by exploiting within-hospital changes in plan-sponsorship status and plausibly exogenous variation in patient assignment to estimate the effect of hospital plan sponsorship on 30-day unplanned readmissions.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting, the data used in the estimation, and the descriptive statistics. Section 3 discusses the sources of endogeneity and the instrumental variables. Section 4 presents evidence using hospital-level data on Fee-for-Service beneficiaries. Section 5 presents the main empirical setting and shows the evidence using MA beneficiaries claims and enrollment data. Section 6 concludes.

2 Institutional Setting and Data

This section describes the institutional environment of Medicare Advantage (MA), the role of provider-sponsored plans, and the data used in the empirical analysis. We first discuss the

growth of vertically integrated provider-sponsored health plans and how these organizations combine payer and provider functions. We then explain why hospital readmissions provide a useful outcome for evaluating the effects of hospital–plan integration. Finally, we describe the administrative datasets used in the analysis and present descriptive statistics on readmission rates and patient episodes across different integration settings.

2.1 Medicare Advantage and Provider-Sponsored plans

Medicare Advantage (MA) is the alternative to traditional Traditional Medicare. Medicare advantage beneficiaries enroll in health plans offered by private insurance companies. Private plans are paid by Medicare a capitated amount per-beneficiary per month to provide coverage for TM Part A and Part B benefits. Plans are differentiated in terms of provider network and coverage, and negotiate with providers the payment rates for each service. Provider-sponsored health plans (PSHPs) are becoming a prominent feature of the Medicare Advantage (MA) market. PSHPs are vertically integrated entities in which a provider organization owns and operates its own insurance plan to provide healthcare services. This integration combines the payer-side (insurance) and provider-side (hospitals, doctors) functions within a single organization. Common ownership may align incentives across payer and provider, reducing transaction costs and enabling clinical integration, potentially raising quality and efficiency. Yet integration can also enable patient steering and increased market power. This vertically integrated organization often takes the form of a health system, but there are instances of health plans that are sponsored by provider organizations that do not include hospitals. In this paper we will exclusively focus on plans sponsored by health systems. We will refer to hospitals that sponsor a plan as vertically integrated (VI) hospitals.

2.2 Incentives for Vertically Integrated Hospitals

This paper focuses on 30-day hospital readmissions as a clinically and financially meaningful outcome through which to study hospital–insurer integration. Let R_i denote an indicator equal to one if patient i experiences an unplanned inpatient readmission within 30 days of discharge, and let

$$q_h \equiv \Pr(R_i = 1 \mid h)$$

denote the probability of readmission following treatment at hospital h . Thirty-day readmissions are common and costly among Medicare beneficiaries and are explicitly targeted

by Medicare’s Hospital Readmissions Reduction Program (HRRP).² Because readmissions are sensitive to discharge planning, transitional care, and post-discharge management, they provide a natural outcome for studying whether changes in hospitals’ financial incentives translate into changes in care delivery.

To illustrate how plan sponsorship changes these incentives, consider first a non-integrated arrangement in which an MA plan reimburses a hospital for an inpatient hospitalization. Hospital payment contracts in Medicare Advantage can take different forms, but existing evidence suggests that inpatient payments commonly remain closely tied to Traditional Medicare reimbursement rates. [Berenson et al. \(2015\)](#), for example, find that MA plans typically pay hospitals rates approximately equal to Traditional Medicare rates, while [Maeda and Nelson \(2018\)](#) similarly find that MA inpatient hospital payments are roughly equal to Medicare fee-for-service payments on average. Traditional Medicare, in turn, reimburses acute inpatient hospitals prospectively on a per-discharge basis, with payments determined primarily by the patient’s MS-DRG.

Motivated by this evidence, let p_h denote the incremental payment received by hospital h from an insurer when an additional inpatient readmission occurs, and let c_h denote the corresponding treatment cost. The hospital’s contribution margin from an additional readmission is

$$m_h = p_h - c_h.$$

If the probability of readmission is q_h , the expected contribution margin associated with readmissions is

$$q_h(p_h - c_h).$$

If $m_h > 0$, an additional readmission generates positive contribution margin for the non-integrated hospital. Absent countervailing incentives such as HRRP penalties or other quality-based payment arrangements, the hospital therefore has no direct financial incentive to reduce readmissions. By contrast, the insurer bears the claims cost associated with the additional hospitalization. When the insurer and hospital are separate organizations, the hospital therefore does not fully internalize the downstream savings generated by preventing rehospitalization.

Plan sponsorship changes this calculation. Let B_i denote the risk-adjusted capitated payment received by an MA plan for beneficiary i . Importantly, B_i does not depend on the cost

²CMS evaluates hospital readmission performance over a multi-year window and applies payment reductions to hospitals with “excess” readmissions relative to national benchmarks. See [Gupta \(2021\)](#).

of care needed to treat patient i . Let $k_h > 0$ denote the cost to the integrated organization associated with an additional readmission. If the readmission occurs at a hospital owned by the same system, k_h reflects the resources required to provide the additional hospitalization. If the readmission occurs at an outside hospital, k_h reflects the payment that the plan must make to the outside provider.

For a beneficiary enrolled in the hospital system's own MA plan, there are two possible states. With probability q_h , the beneficiary is readmitted and the integrated organization receives the capitated payment B_i but incurs the downstream cost k_h . With probability $1 - q_h$, the beneficiary is not readmitted and the organization receives B_i without incurring this additional cost. Expected surplus can therefore be written as

$$\Pi_h^{VI} = q_h(B_i - k_h) + (1 - q_h)B_i.$$

Simplifying,

$$\Pi_h^{VI} = B_i - q_h k_h.$$

Hospitals may respond to this incentive by investing in care-management capacity. Let e_h denote investment in activities such as discharge planning, care-management staffing, timely outpatient follow-up, medication reconciliation, post-discharge monitoring, and coordination with outpatient and post-acute providers. Suppose that investment lowers the probability of readmission,

$$q_h = q(e_h), \quad q'(e_h) < 0,$$

and let $F(e_h)$ denote the cost of investment. Expected surplus for an own-plan beneficiary can then be written as

$$\Pi_h^{VI}(e_h) = B_i - q(e_h)k_h - F(e_h).$$

Because B_i does not depend on e_h , the marginal return to investment for the hospital system's own-plan beneficiaries is

$$\frac{\partial \Pi_h^{VI}(e_h)}{\partial e_h} = -q'(e_h)k_h - F'(e_h).$$

Thus, plan sponsorship increases the return to investments that reduce readmissions among beneficiaries enrolled in the hospital system's own plan: when investment lowers $q(e_h)$, the integrated organization avoids downstream costs that would otherwise be borne by the insurer.

Importantly, many investments in care management may operate at the level of the hospital or clinical practice rather than being targeted exclusively to patients enrolled in a particular insurance plan. Changes in discharge protocols, care-management staffing, information systems, medication reconciliation, or coordination with post-acute providers may therefore affect the treatment of patients across payers. This possibility is consistent with evidence that managed-care incentives can generate spillovers to patients outside the plans directly subject to those incentives (Glied and Zivin, 2002; Baicker et al., 2013). Plan sponsorship can therefore create incentives originating from the hospital system’s own insured population while generating changes in care for a broader set of patients treated at the hospital.

At the same time, such hospital-wide spillovers introduce an additional tradeoff. Preventing a readmission for a patient insured by an outside plan may cause the hospital to forgo the contribution margin that it would otherwise earn from that admission. Whether the financial benefit generated by the hospital’s own-plan population is sufficiently large to induce greater hospital-wide investment therefore depends on the composition of patients and on the provider margin associated with admissions of patients insured by outside plans. We formalize this tradeoff in the next subsection and show how it varies with hospital market concentration.

2.3 Hospital Market Concentration and Vertically Integrated Hospitals

Hospital market structure may shape the extent to which vertical integration changes hospitals’ incentives to invest in reducing readmissions. The key mechanism arises when investments in care management operate at the hospital level and therefore affect patients enrolled both in the hospital system’s own Medicare Advantage plan and in outside insurance plans.

Let α_h denote the share of patients at hospital h who are enrolled in the hospital system’s own MA plan. As above, let e_h denote hospital-wide investment in care management and let the probability of readmission be $q(e_h)$, where

$$q'(e_h) < 0.$$

For patients enrolled in the hospital’s own MA plan, a readmission creates a downstream cost for the integrated organization. Let $k_h > 0$ denote the cost internalized by the integrated system when one of its own-plan patients is readmitted. Preventing such a readmission therefore generates a financial benefit of k_h to the integrated organization.

The incentives are different for patients insured by outside plans. Let p_h denote the pay-

ment received by the hospital from an outside insurer for an additional inpatient admission, and let c_h denote the corresponding treatment cost. The hospital's contribution margin from a readmission of an outside-plan patient is

$$m_h = p_h - c_h.$$

When $m_h > 0$, preventing a readmission for an outside-plan patient causes the hospital to forgo this contribution margin. Hospital-wide investments therefore generate two opposing financial effects for an integrated system: they reduce downstream spending for patients enrolled in the system's own plan, but they may also reduce profitable admissions among patients insured by outside plans.

Let $F(e_h)$ denote the cost of hospital-wide care-management investment. The portion of the integrated organization's expected profit that depends on readmissions and investment can be written as

$$\Pi_h^{VI}(e_h) = -\alpha_h q(e_h) k_h + (1 - \alpha_h) q(e_h) m_h - F(e_h).$$

where α_h denote the share of patients treated at hospital h who are enrolled in the hospital system's own Medicare Advantage plan. The first term captures the expected downstream cost of readmissions among the hospital system's own plan enrollees. The second captures the contribution margin generated by readmissions among patients insured by outside plans. Differentiating with respect to investment gives

$$\frac{\partial \Pi_h^{VI}}{\partial e_h} = -q'(e_h) [\alpha_h k_h - (1 - \alpha_h) m_h] - F'(e_h).$$

Thus, the net marginal financial benefit from reducing readmissions is

$$\alpha_h k_h - (1 - \alpha_h) m_h.$$

The first component, $\alpha_h k_h$, represents the savings generated by preventing downstream utilization among the integrated system's own enrollees. The second component, $(1 - \alpha_h) m_h$, represents the provider margin forgone when the same hospital-wide investment prevents readmissions among patients insured by outside plans. A strictly positive level of investment is optimal when the marginal benefit of reducing readmissions is sufficiently large to exceed the marginal cost of investment. In particular, a sufficient condition for positive investment is

$$-q'(0) [\alpha_h k_h - (1 - \alpha_h) m_h] > F'(0).$$

Hospital market concentration affects this tradeoff through the provider-side margin m_h . Hospitals in more concentrated markets have greater bargaining power vis-à-vis insurers and may therefore obtain higher negotiated prices. Holding treatment costs fixed,

$$p_h^C > p_h^U$$

implies

$$m_h^C > m_h^U,$$

where C and U denote concentrated and unconcentrated hospital markets, respectively. Consequently,

$$\alpha_h k_h - (1 - \alpha_h) m_h^C < \alpha_h k_h - (1 - \alpha_h) m_h^U.$$

The marginal return to hospital-wide investments that reduce readmissions is therefore lower in concentrated hospital markets. Under standard assumptions that investment costs are convex and that the marginal effectiveness of care-management investment diminishes with investment, and holding α_h , k_h , and the technology $q(e_h)$ fixed across markets, this implies

$$e_h^C < e_h^U.$$

The intuition is that vertical integration creates an incentive to reduce readmissions among the hospital system's own plan enrollees because the system internalizes the resulting downstream savings. However, when the resulting care-management investments also reduce readmissions among patients insured by outside plans, the hospital gives up the contribution margin associated with those admissions. This opportunity cost is larger when negotiated hospital prices are high. Consequently, the additional incentive created by plan sponsorship to make hospital-wide investments in reducing readmissions is stronger in more competitive hospital markets, where the provider-side margin forgone when an admission is prevented is smaller.

This mechanism therefore generates the empirical prediction that the reduction in readmissions associated with hospital plan sponsorship should be larger in unconcentrated hospital markets than in concentrated hospital markets.

2.4 Data

Medicare Advantage Claims Data We use administrative data from the Centers for Medicare & Medicaid Services (CMS) for the years 2019 and 2022. The main data source for plan enrollment is the Master Beneficiary Summary File (MBSF), which contains individual-level data on a 5% random sample of Medicare beneficiaries. The MBSF records enrollment information for each beneficiary, including whether they are enrolled in Traditional Medicare or a Medicare Advantage (MA) plan.

The MBSF also includes detailed beneficiary characteristics such as age, gender, race, income measures, and county of residence. In 2019, total Medicare enrollment was approximately 61 million, so the 5% sample includes about 3 million individuals. Roughly 37% of beneficiaries were enrolled in MA plans. We exclude ESRD patients from the analysis and also special needs plans, pace plans, employer-sponsored plans or PFFS plans which have a different reimbursement scheme from CMS which is different from the standard plans.

The main data source for patients’ acute-care utilization is the Inpatient Encounter Data. These files identify the admitting hospital (CCN) and, for ambulance-transported episodes, the ambulance provider. They include ICD-10 diagnosis and procedure codes, length of stay, and discharge disposition, which we use to construct episode definitions, risk adjusters, and readmission outcomes. In addition, we use the Inpatient, Outpatient, Skilled Nursing Facility (SNF), Home Health Agency (HHA), Durable Medical Equipment (DME), and Carrier Encounter files to measure broader utilization and to construct spending-based measures in supplementary analyses.

Provider-sponsored health plans and VI hospitals. To identify integrated MA plans sponsored by health systems, we use the AHRQ Compendium of Health Systems, which contains the list of all health plans offered by hospitals and health systems. We link the hospitals in the Compendium to their CMS provider identification numbers (CCN) in order to merge the VI classification to hospital-level datasets (including the AHA Annual Survey for hospital characteristics). To identify health plans sponsored by other provider organizations, we additionally search through the full list of MA contracts and flag those whose sponsor is a provider organization. Throughout the descriptive analysis, we classify a hospital as vertically integrated in a given year if it sponsors at least one plan in that year. In the empirical section, we also consider a more robust definition of hospital VI integration status based on the number of beneficiaries enrolled in the hospital-plan relative to the capacity of the hospital.

Readmission measure We use the claims data to compute readmission at the individual level. A patient experiences an *unplanned* acute inpatient readmission if within 30 days of discharge from the *index* emergency admission, the patient is readmitted to an hospital for the same condition. We treat multiple rehospitalizations within the 30-day window as a single event, consistent with HRRP-style targeted measures, which are most naturally interpreted as readmission probabilities rather than counts.³ Following CMS-style measurement conventions, we exclude same-day transfers and implement a planned-readmission algorithm to remove rehospitalizations that are predictably scheduled (e.g., selected elective procedures) rather than reflecting failures of post-discharge management.⁴ Because our data are Medicare Advantage encounters, we can track readmissions to *any* acute-care hospital, not only the index hospital.

2.5 Summary Statistics

We begin by presenting descriptive statistics for both the estimation sample and the full inpatient sample. Our estimation sample is restricted to the emergency episodes and is the one used in the hospital-side instrumental-variables design, while the broader full inpatient sample is useful for assessing how restrictive that sample selection is in practice. Across both samples, the raw readmission patterns are remarkably similar: the overall 30-day unplanned readmission rate is 14.5 percent in the emergency sample and 14.5 percent in the full sample. This comparison is useful because it shows that the emergency restriction does not mechanically generate an unusual readmission environment.

Readmission Rates [Table 1](#) reports readmission comparisons across levels of hospital market concentration. Raw readmission rates are highest in unconcentrated markets, lower in moderately concentrated markets, and lowest in highly concentrated markets. The raw association between admission to a vertically integrated hospital and readmission is not stable across market structures. In unconcentrated markets, readmission rates are slightly higher at VI hospitals than at non-VI hospitals, whereas in highly concentrated markets they are slightly lower. This instability in the unconditional hospital-side comparison suggests that simple raw differences are unlikely to isolate the causal effect of hospital integration. The empirical section therefore turns to an IV design that addresses patient sorting into hospitals.

³[Gupta \(2021\)](#) notes that CMS does not differentiate between one and multiple readmissions within the 30-day period when constructing targeted readmission metrics.

⁴[Ziedan and Kaestner \(2021\)](#) describe the 30-day all-cause framework and common exclusions used in HRRP-style definitions.

Table 1: 30-Day Unplanned Readmission by Hospital Integration and Market Concentration

Market Concentration	Hospital Integration	
	Non-VI	VI
<i>Emergency episode sample</i>		
HHI < 1500	15.4	15.9
$1500 \leq \text{HHI} < 2500$	15.1	15.1
HHI ≥ 2500	14.1	13.7
Total	14.5	14.5
<i>Full inpatient sample</i>		
HHI < 1500	15.4	15.9
$1500 \leq \text{HHI} < 2500$	15.1	15.0
HHI ≥ 2500	14.1	13.7
Total	14.5	14.5

Notes: Each cell reports the mean 30-day all-cause unplanned readmission rate (in percent). Hospitals are classified as vertically integrated (VI) if they sponsor a Medicare Advantage plan. Market concentration is measured using county-level hospital HHI. The Total row reports mean readmission rates across all levels of market concentration.

Distribution of Patient Admissions Table 2 reports how admissions are distributed across vertically integrated and non-integrated hospitals. In the emergency episode sample, 35.8% of admissions occur at vertically integrated hospitals, while 64.2% occur at non-integrated hospitals. In contrast, vertically integrated hospitals account for only 7.4% of admissions in the full inpatient sample, with the remaining 92.6% occurring at non-integrated hospitals.

The substantially higher share of admissions to vertically integrated hospitals in the emergency episode sample highlights an important difference in sample composition. Although the emergency and full inpatient samples exhibit similar unconditional readmission rates, they differ considerably in the types of hospitals at which patients receive care. This distinction is important when interpreting the emergency-based causal estimates and suggests caution in extrapolating those estimates to the broader population of inpatient admissions.

Sample restriction, limitations, and generalizability The hospital-side instrumental-variables design relies on ambulance referral patterns, so the main estimation sample is restricted to high-acuity, effectively non-deferrable emergency admissions. This follows the logic in Doyle Jr et al. (2015) and related ambulance-based designs: the identifying assumption is most credible when pre-hospital assignment and ambulance routing, rather than

Table 2: Distribution of Admissions by Hospital Vertical Integration

Sample	Hospital Integration		Total
	Non-VI	VI	
<i>Emergency episode sample</i>	137,524	76,725	214,249
	64.19%	35.81%	100.00%
<i>Full inpatient sample</i>	969,175	77,022	1,046,197
	92.64%	7.36%	100.00%

Notes: Each cell reports the number of admissions and its percentage of the sample total. Non-VI hospitals are hospitals that do not sponsor a Medicare Advantage plan. VI hospitals are hospitals that sponsor a Medicare Advantage plan.

patient discretion, largely determine the admitting hospital. Subsequent work using the same design likewise focuses on emergency settings for exactly this reason (Cooper et al., 2022). The advantage of the restriction is internal validity. By concentrating on acute admissions for which location choice is least discretionary, the ambulance-based instrument is more plausibly orthogonal to patient unobservables.

The tradeoff is external validity. The hospital-side estimates should be interpreted as applying most directly to acute emergency inpatient episodes rather than to elective admissions, chronic disease management, or the full universe of inpatient utilization. The descriptive evidence is informative about the scope of this limitation. On the one hand, Table 1 shows that overall readmission levels and the raw plan-versus-hospital patterns are nearly unchanged when moving from the emergency sample to the broader inpatient sample. On the other hand, Table 2 show that the distribution of admissions across integrated settings differs more materially in the broader sample, especially for beneficiaries enrolled in vertically integrated plans. Accordingly, the emergency restriction should be viewed as a design choice that strengthens identification, not as a claim that the same treatment effects must hold uniformly for all hospital admissions.

3 Endogeneity and Identification Strategy

There are two sources of endogeneity concerns when estimating the effect of hospital-plan integration on readmission rates. We refer to the first as supply-side selection. Hospitals and plans may differ systematically in unobservable quality which also affects the readmission rate. For example, high-quality hospitals plans may be more likely to sponsor health plans compared to low-quality hospitals. The second concern is patient selection. Patients with ex-ante different readmission probability may be more or less likely to enroll in PSHP or

being treated by plan-sponsor hospitals. In this section, we discuss how we address both of these challenges.

3.1 Supply side endogeneity and identification

Hospital side To address hospital selection into plan-sponsoring, we need to control for hospital level unobserved quality. We use hospital fixed effects to absorb all time-invariant hospital attributes, including baseline quality, discharge practices, coding intensity, and persistent post-acute referral patterns. County fixed effects absorb time-invariant local conditions that may affect both hospital choice and readmission risk. Year and diagnosis fixed effects absorb common shocks and differences in clinical composition.

Under the standard parallel trend assumption, we can identify the impact of VI hospital in a difference in difference design with hospital and year fixed effects. In this setting identification comes from within-hospital changes in plan integration status over time, net of aggregate time trends. [Table 3](#) shows one of our key identifying variations. It summarizes the distribution of hospitals by their vertical integration status over time. A majority of hospitals (64%) are never vertically integrated during the sample period, while 20% remain vertically integrated throughout. Importantly, 16% of hospitals switch their vertical integration status at least once over time. This group of “switchers” provides the key source of identification in our difference-in-differences framework with hospital and year fixed effects, as it reflects within-hospital changes in integration status.

Table 3: Hospital Vertical Integration Status Over Time

	Frequency	Percentage (%)
Always VI	555	19.97
Always non-VI	1,776	63.91
Switchers	448	16.12
Total Hospitals	2,779	100.00

However, patient selection may still pose a concern if the composition of patients changes in response to plan-integration. In particular, if healthier patients differentially sort into vertically integrated settings following changes in integration status, estimates may remain biased. To address this concern, we complement our difference-in-differences strategy with an instrumental variable approach. In section 3.2, we describe instrument we use to account for patient selection into hospitals.

Another endogeneity concern is that hospitals may adopt VI status as part of a broader organizational change—for example, improvements in care management, discharge planning, or system integration—that begins before formal adoption. In [subsection 4.2](#) we present a

validation exercise using hospital-level fee-for-service (FFS) hospital-episode-year data from 2016–2023. The scope of this exercise is to check whether plan-sponsor hospitals and standard hospitals were on different readmission trends prior to the hospital’s VI adoption event.

3.2 Instruments for Patient Selection

This section presents the instruments we use to address the possibly non-random selection of beneficiaries between hospitals.

Ambulance-based instrument for hospital assignment In general, patients’ choice over hospitals depends on unobserved hospital quality which might also be correlated with readmission probability. However, in emergency episodes, the admitting hospital is heavily influenced by pre-hospital factors such as ambulance dispatch, ambulance availability, and patient location. Patients typically do not choose the ambulance company, and dispatch/rotation protocols together with mutual-aid agreements generate plausibly exogenous assignment to ambulance companies with heterogeneous hospital destination preferences (Doyle Jr et al., 2015). At the same time, ambulance companies often have stable tendencies to deliver patients to particular hospitals, reflecting historical routing patterns, contracts, operational convenience, or local familiarity.

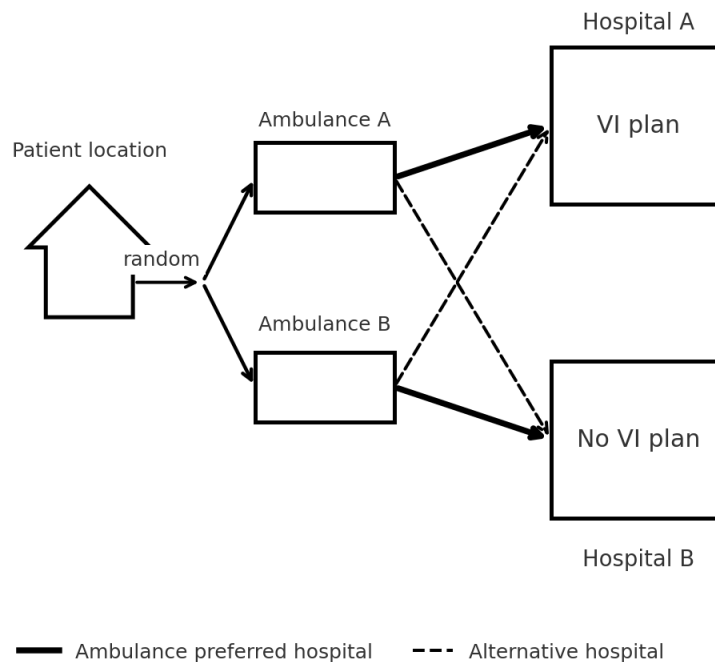


Figure 1: Hospital Assignment and Ambulance Preferences

Figure 1 illustrates the key source of quasi-random variation in hospital assignment. Two ambulances serve the same geographic area but differ in their propensity to transport patients to different hospitals. If one ambulance is systematically more likely to deliver patients to a VI hospital than another, then patients who are quasi-randomly assigned to those ambulances differ in their probability of admission to a VI hospital even though they face similar emergencies and originate from the same local area.

Let H_i be the indicator for *admission to a vertically integrated hospital* for beneficiary-episode i , and let $a(i)$ identify the ambulance company serving episode i for diagnosis category $d(i)$. We construct the leave-one-out ambulance-level VI-hospital share:

$$Z_i^{A,H} = \frac{1}{N_{a(i),d(i),t(i)} - 1} \sum_{j: \substack{a(j)=a(i) \\ d(j)=d(i)}} H_j.$$

Thus, $Z_i^{A,H}$ is the leave-one-out share of episodes with the same ambulance company, diagnosis group, and year that are admitted to a VI hospital. Conditioning on diagnosis is useful because ambulance routing patterns may differ across emergency conditions.

The identifying intuition is that, among otherwise similar patients facing similar emergencies, assignment to an ambulance with a higher propensity to transport patients to VI hospitals raises the probability of admission to a VI hospital while remaining plausibly orthogonal to patient unobservables conditional on location and the control set.

Differential distance as a common geographic shifter We additionally use a geographic shifter based on relative proximity to VI and non-VI hospitals. Let $\text{dist}(r(i), h)$ denote distance from beneficiary i 's residence (or pickup location) $r(i)$ to hospital h . Define the nearest VI hospital $h^{VI}(i)$ and the nearest non-VI hospital $h^{NVI}(i)$, and construct

$$Z_i^{DD} \equiv \ln \text{dist}(r(i), h^{VI}(i)) - \ln \text{dist}(r(i), h^{NVI}(i)).$$

More negative values of Z_i^{DD} indicate greater relative proximity to a VI hospital.

This same differential-distance measure is useful for both endogenous margins, but with different interpretations. First, in the hospital-assignment equation, Z_i^{DD} shifts the probability that an emergency episode is delivered to a VI hospital, because distance is a first-order determinant of hospital destination in emergency care. Second, in the plan-enrollment equation, Z_i^{DD} shifts the relative attractiveness of enrolling in a provider-sponsored plan, because beneficiaries value proximity to in-network hospitals and providers. In that setting, proximity to a VI hospital system makes the affiliated plan more attractive. Thus, the same geographic shifter enters both the hospital and plan instrument sets, but its identifying

role differs depending on whether the endogenous regressor is hospital assignment or plan enrollment.

3.3 Excluded instruments

Combining the margins above, the excluded instrument set for admission to a VI hospital is

$$Z_i^H \equiv (Z_i^{A,H}, Z_i^{DD}).$$

The first component captures ambulance-specific destination preferences; the second captures relative geographic proximity to VI versus non-VI hospitals. The identifying assumption is that, conditional on the control set and fixed effects, the ambulance preference shifter and differential distance affect readmissions only through admission to a VI hospital. This assumption is most plausible in high-acuity emergency episodes, where patient discretion over the timing and location of care is limited.

A common stability assumption applies to the ambulance-based instruments used in this paper: conditional on diagnosis, geography, and the fixed effects, ambulance companies' destination preferences do not themselves change in response to a hospital becoming a plan sponsor in a way that is directly related to post-discharge outcomes. Put differently, hospital adoption of plan sponsorship do not induce ambulance providers to re-optimize routing toward that hospital for reasons other than the treatment margin being instrumented. This assumption is most plausible in high-acuity emergency episodes, where ambulance destination patterns are shaped primarily by operational protocols, capacity, and geography rather than by insurer ownership.

The excluded instrument set for enrollment in a VI plan is

$$Z_i^P \equiv (Z_i^{A,P}, Z_i^{DD}).$$

The first component captures county-level PSHP availability; the second captures the relative convenience of proximity to a VI hospital system. For the plan-enrollment equation, the identifying assumption is that, conditional on beneficiary controls, plan characteristics, fixed effects, and admitting hospital, county-level PSHP availability and differential distance affect readmissions only through VI-plan enrollment. The hospital fixed effects in the plan regression are particularly important: they ensure that identification does not come from VI-plan enrollees being treated at systematically different hospitals on average. Rather, the design compares beneficiaries treated at the same hospital but facing different plan-enrollment incentives.

3.4 Estimation samples

For the hospital-treatment margin, we restrict attention to high-acuity, effectively non-deferrable emergency conditions—such as acute myocardial infarction, stroke, and hip fracture—for which the index admission is largely outside the patient’s control. This sharpens the plausibility of the ambulance-based assignment design.

The emergency estimation sample spans a broad set of diagnosis categories. The most common groups are circulatory-system conditions (23.25%), infectious diseases (17.12%), injuries and trauma (13.57%), respiratory-system conditions (10.77%), and digestive-system conditions (8.86%). Additional episodes come from genitourinary conditions (6.87%), endocrine disorders (4.48%), and nervous-system conditions (4.00%), with the remaining 11.08% classified as other diagnoses. Thus, while the empirical design is built around emergency and non-deferrable admissions, the estimation sample is not limited to a single clinical condition; rather, it covers a broad range of acute diagnoses for which readmission is a meaningful indicator of inpatient quality and post-discharge care coordination.

4 Evidence from Hospital-level FFS data

Before turning to MA beneficiary-level analysis, we present evidence using a panel dataset on hospital-level for Fee-for-Service (FFS) beneficiaries from 2013 to 2023. We implement the staggered difference-in-difference design to estimate the impact of hospital VI status on readmission rate for different admission diagnosis following [Callaway and Sant’Anna \(2021\)](#). This empirical framework has two main strengths. First, since FFS beneficiaries are covered by Traditional Medicare, endogeneity threat related to plan side selection does not apply. Secondly, having a longer panel allows us to both exploit more variation in hospital VI status over time, and also to test for differential pre-trends in readmission rates among VI hospitals and standard hospitals.

However, this design has two important limitations. First, aggregate hospital level data do not allow us to control for unobservable patient selection across hospital. Secondly, the introduction of a sponsored plan may still indirectly affect the composition of the observed FFS population through switching between Traditional Medicare and Medicare Advantage or through broader changes in referral patterns and system organization. For this reasons, we consider this analysis as complementary to that of Section 5 where we use patient level data and control for patient selection using the IV design discussed in Section 3. A key assumption that applies to both the DiD analysis and the IV framework is that the timing of the hospital VI-status adoption is exogenous. In other words, we cannot rule out that hospital

VI adoption may coincide with broader organizational changes—for example, improvements in care management, discharge planning, or system integration—that affect readmissions independently of plan sponsorship itself.

4.1 Staggered Difference-in-differences design

Let h index hospitals, e episode categories, $m(h)$ the hospital’s local market, and t year. Let RR_{het} is the observed 30-day unplanned readmission rate for hospital h , episode e , and year t ; VI_{ht} indicates whether hospital h sponsors a Medicare Advantage plan in year t . We implement a simple Difference-in-Difference test to test whether there is evidence of hospital-plan integration affecting the readmission rates among FFS beneficiaries. We run two separate specifications: one in the full FFS sample and one in the subsample of unconcentrated hospital markets, defined as county-year markets with hospital HHI below 1500. In the baseline specification, we estimate regressions of the form

$$\ln(RR_{het}) = \beta VI_{ht} + \rho \ln(ERR_{het}) + X'_{ht}\gamma + Z'_{m(h)t}\delta + \alpha_h + \nu_{r(h)} + \eta_e + \tau_t + \epsilon_{het}, \quad (1)$$

where $\ln(ERR_{het})$ is the log expected 30-day readmission rate for hospital h , episode e , and year t . This is useful because hospitals differ over time in the underlying risk profile of the patients they treat. ERR_{het} provides a compact case-mix-adjusted benchmark for that cell, so conditioning on $\ln(ERR_{het})$ helps ensure that the estimated effect of VI adoption is not simply capturing predictable variation in readmissions driven by changes in patient severity or episode composition. The other controls are: X_{ht} denotes time-varying hospital controls; $Z_{m(h)t}$ denotes time-varying market controls; α_h are hospital fixed effects; $\nu_{r(h)}$ are HRR fixed effects; η_e are episode fixed effects; and τ_t are year fixed effects. Regressions are weighted by $\ln(\text{discharges}_{het})$, and standard errors are clustered at the hospital level.

Table 4 reports the Callaway and Sant’Anna average treatment effect on the treated (CSDID ATT) results using FFS hospital–episode–year data, where the dependent variable is the log 30-day unplanned readmission rate. In the full sample, hospital VI adoption is not associated with a statistically significant decline in readmissions. We also estimate the specification in Equation 1 for the subset of hospitals that operate less concentrated markets ($\text{HHI} < 1500$). In unconcentrated markets, the coefficient of interest is statistically significant and indicate a lower readmission rate of about 2.6 percent. These patterns are consistent with the hospital-side results from the IV analysis and suggest that the effects of vertical integration may be stronger in less concentrated markets. Because the analysis is conducted on FFS patients, the estimates are more naturally interpreted as reflecting

hospital-wide changes in care delivery that apply to all patient treated at VI hospitals.

Table 4: Impact of Hospital VI Adoption on FFS Readmission Rates

	Full Sample	Unconcentrated Markets	Concentrated Markets
	CSDID ATT	CSDID ATT	CSDID ATT
Hospital VI adoption	-0.0077 (0.0079)	-0.0263** (0.0120)	-0.0026 (0.0100)
Observations	84,368	46,479	37,444
Hospital FE	Yes	Yes	Yes
HRR FE	Yes	Yes	Yes
Episode FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Hospital controls	Yes	Yes	Yes
Market controls	Yes	Yes	Yes

Notes: The dependent variable is the log 30-day unplanned readmission rate at the hospital-episode-year level. Hospital VI adoption equals one if the hospital sponsors a Medicare Advantage plan in year t . Estimates report the Callaway and Sant’Anna average treatment effect on the treated (CSDID ATT). All specifications control for the log expected readmission rate, time-varying hospital characteristics, and time-varying market characteristics. Unconcentrated markets are county-year hospital markets with HHI < 1500 , while concentrated markets have HHI ≥ 1500 . Regressions are weighted by $\ln(\text{discharges})$, and standard errors clustered at the hospital level are reported in parentheses.

4.2 Pre-trend assessment

A key identifying assumption of the difference-in-differences design is that, absent adoption, hospitals that later become vertically integrated would not already have been on systematically different readmission trajectories. For example, hospitals that later become plan-sponsor could have already implemented improvement in care management that allowed to reduce readmission risk. In this case, our DiD estimation can be biased because lower readmission rates at hospitals observed as vertically integrated could partly reflect pre-existing trends rather than the effect of VI adoption itself. This concern applies not only to the DiD design but also to the IV design in the next section. However, since we only have two years of MA beneficiary-level data we cannot test for the absence of pre-trends in that case. Under the assumption that VI hospitals implement hospital-wide change in care delivery, we consider the pre-trend assessment using FFS data to be informative also for the MA populations.

To assess this threat, we estimate an event-study specification centered on each hospital’s *first* observed year of VI adoption. Let G_h denote the first year hospital h becomes vertically

integrated, and define event time as $k = t - G_h$. We then estimate:

$$\begin{aligned} \ln(RR_{het}) = & \sum_{k \neq -1} \theta_k \mathbb{1}\{t - G_h = k\} \\ & + \beta \ln(ERR_{het}) + X'_{ht}\gamma + Z'_{m(h)t}\delta \\ & + \alpha_{he} + \lambda_{et} + u_{het}, \end{aligned} \tag{2}$$

where RR_{het} is the observed FFS readmission rate for hospital h , episode e , and year t ; ERR_{het} is the expected readmission rate for that hospital-episode-year cell; X_{ht} denotes time-varying hospital controls; $Z_{m(h)t}$ denotes time-varying market controls; α_{he} are hospital-episode fixed effects; and λ_{et} are episode-year fixed effects. Standard errors are clustered at the hospital level. The omitted event-time category is $k = -1$, so each θ_k measures the difference in log readmission rates at event time k relative to the year immediately before first adoption.

In the event study design specified in [Equation 2](#), hospital-episode fixed effects absorb all time-invariant differences across hospital-episode cells, while episode-year fixed effects absorb common all time-varying episode-specific shocks. Conditional on expected risk, hospital controls, market controls, and these high-dimensional fixed effects, the lead coefficients θ_k for $k < 0$ provide a direct test of whether hospitals that later adopt VI were already on different outcome trajectories before adoption.

[Figure 2](#) plots the estimated event-time coefficients for the *unconcentrated market*. The pre-adoption coefficients are small in magnitude and statistically imprecise. In the baseline event-study specification, the coefficients at $k = -3$ and $k = -2$ are approximately 0.0087 and -0.0007 , respectively, with neither individually distinguishable from zero.

However, a joint test of the pre-treatment coefficients rejects the null that they are jointly zero ($\chi^2 = 80.38$, $p < 0.001$). This suggests some deviation from parallel trends in the pre-adoption period, even though individual coefficients are imprecisely estimated.

The post-adoption coefficients are negative and become more pronounced over time. The coefficient at $k = 0$ is close to zero, while the estimates at $k = 1$, $k = 2$, and $k = 3$ are negative, with the largest decline appearing three years after first adoption. In the current specification, the estimate at $k = 3$ is approximately -0.042 and statistically significant at conventional levels.

Because the purpose of this exercise is primarily to evaluate pre-trends, we do not emphasize these lag coefficients as separate causal treatment effects. Nevertheless, their sign is consistent with lower readmission rates after adoption.

[Appendix A](#) reports the corresponding event-study results for the full sample and concentrated markets.

Taken together, this evidence yields two conclusions. First, within a hospital, VI adoption is associated with lower readmission rates in unconcentrated markets. Second, the event-study evidence provides no indication that future adopters were already on differential pre-adoption readmission trajectories in the observed pre-period. These findings strengthen the hospital-adoption margin as a useful benchmark and weaken the concern that the main hospital-side results are driven entirely by pre-existing differential trends.

In the next section, we implement a difference-in-difference design using MA beneficiary level data. These data allow us to implement an IV strategy to address endogeneity from patient selection across hospitals, but also introduce another source of endogeneity that stems from unobservable plan-side effects.

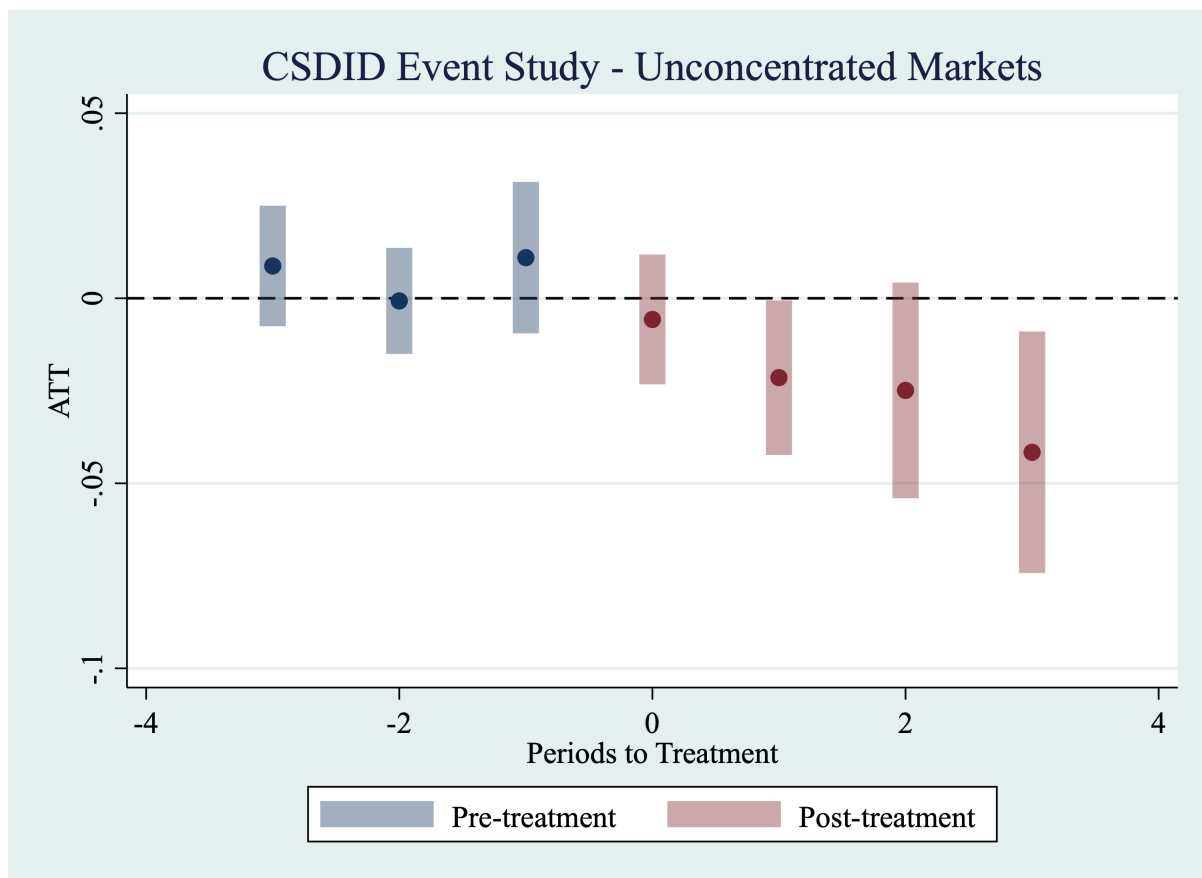


Figure 2: FFS event study around first hospital VI adoption

5 Evidence from beneficiary level MA data

The unit of observation is the beneficiary-by-episode, where an episode is defined as an inpatient admission originating in the emergency department (or an observation stay subse-

quently converted to inpatient) with no transfer-in from another acute-care facility. For each episode, we observe the beneficiary’s Medicare Advantage (MA) plan enrollment, the admitting hospital, and whether the beneficiary experiences an unplanned readmission within 30 days for any cause.

Let i index the beneficiary-episode. The primary outcome is an indicator for 30-day all-cause unplanned readmission,

$$R_i = \begin{cases} 1 & \text{if beneficiary } i \text{ has an unplanned readmission within 30 days,} \\ 0 & \text{otherwise.} \end{cases}$$

Let $h(i)$ denote the admitting hospital, $p(i)$ the MA plan, and $t(i)$ the year. We study three treatment margins: (i) admission to a vertically integrated hospital, (ii) enrollment in a vertically integrated plan, and (iii) *full integration*, meaning enrollment in a VI plan and treatment at the sponsoring hospital. Define

$$H_i \equiv \mathbb{1}\{\text{beneficiary } i \text{ is admitted to a VI hospital in year } t(i)\},$$

$$P_i \equiv \mathbb{1}\{\text{beneficiary } i \text{ is enrolled in a VI plan in year } t(i)\},$$

and

$$F_i \equiv \mathbb{1}\{\text{beneficiary } i \text{ is enrolled in a VI plan and treated at the sponsoring hospital}\}.$$

Across all three treatment margins, we report specifications in two samples: the full estimation sample and the subsample of beneficiary episodes treated at hospitals operating in unconcentrated hospital markets, defined as county-year markets with $HHI < 1500$. The reason is that if vertical integration affects hospital-wide discharge planning, care coordination, or post-acute management, those responses may be stronger in less concentrated provider markets, where competitive pressure is greater.

LATE interpretation. The IV hospital estimates identify a local average treatment effect for beneficiaries whose hospital assignment is shifted by ambulance referral patterns and differential distance. Because the hospital specifications also include hospital fixed effects, they combine a complier interpretation on the patient-assignment margin with within-hospital variation over time in VI status.

The VI-plan estimates identify a local effect of instrument-induced enrollment into VI plans, conditional on the admitting hospital and observed plan characteristics. Because plan fixed effects are infeasible, those estimates should not be interpreted as within-plan

ownership effects.

The full-integration estimates identify the additional effect of sponsor-plan enrollment among beneficiaries whose fully integrated treatment status is shifted by the product-based instrument, but those specifications turn out to be weakly identified in the data and are therefore interpreted cautiously.

5.1 Admission to a VI hospital

We begin with admission to a vertically integrated hospital. The baseline outcome is either an indicator for 30-day unplanned readmission or length of stay (in logs). As noted above, we report each hospital-side specification in both the full sample and the subsample of unconcentrated hospital markets. We run the following specification:

$$Y_i = \beta_H H_i + X_i' \gamma + \alpha_h + \phi_p + \mu_c + \lambda_d + \tau_{t(i)} + \varepsilon_i, \quad (3)$$

where H_i indicates admission to a vertically integrated hospital, X_i contains beneficiary controls, α_h are hospital fixed effects, ϕ_p are plan fixed effects, μ_c are county fixed effects, λ_d are diagnosis fixed effects, and $\tau_{t(i)}$ are year fixed effects. The specification in [Equation 3](#) does not have a time subscript because most beneficiaries appears only once in our sample. The hospital analysis uses the emergency-episode sample in which ambulance-based assignment shifters are available. Patient-side selection is addressed by instrumenting H_i with the leave-one-out ambulance preference shifter $Z_i^{A,H}$ and differential distance Z_i^{DD} . Hospital and plan fixed effects absorb all time invariant difference across hospitals and plans respectively. Under the assumption that the timing of hospital adoption of VI status is exogenous, the coefficient of interest, β_H , admits a causal interpretation.

[Appendix B](#) reports robustness analysis based on a stricter VI-hospital definition. In that alternative definition, a hospital-year is treated as vertically integrated only when sponsor-plan enrollment relative to bed capacity is above the median. The motivation is hospital-wide incentives: if the mechanism operates through organizational changes in discharge planning and care coordination, those changes should be more likely when the hospital has a larger mass of patients for whom it fully internalizes the gains from avoiding readmissions. We find that results are robust to this definition.

5.2 Full integration

$$Y_i = \beta_F F_i + \beta_H H_i + X_i' \gamma + \alpha_h + \phi_p + \mu_c + \lambda_d + \tau_{t(i)} + \varepsilon_i, \quad (4)$$

where F_i indicates full integration: the beneficiary is enrolled in a VI plan and is treated

at the hospital that sponsors that plan. The full-integration specification asks whether there is an additional effect of sponsor-plan enrollment once the patient is already treated at a VI hospital. While the overall omitted group includes all patients with $F_i = 0$, the coefficient on F_i is identified by comparing fully integrated patients to those treated at the same VI hospitals but not enrolled in the sponsoring plan. We instrument F_i using the interaction-based shifter, which is given by the product of the hospital-assignment and plan-enrollment instruments already described in Section 4.

Conceptually, this is the test of within-hospital differential treatment by plan ownership. If fully integrated patients performed better even relative to other patients treated at the same VI hospital, that would suggest that VI hospitals manage sponsor-plan enrollees differently from other patients. If not, the hospital-side effect is more naturally interpreted as a hospital-wide organizational response that benefits all patients treated at the VI hospital.

5.3 Results

Hospital Side Results Table 5 reports the results of the baseline hospital-side specification in Equation 3. The OLS and IV estimates point in very different directions. In the full sample, OLS implies that patients treated at VI hospitals have higher readmission rates, whereas the IV estimate is close to zero and statistically insignificant. We interpret that gap as evidence of adverse patient selection into VI hospitals. Once the sample is restricted to unconcentrated markets, however, the IV coefficient becomes negative and statistically significant. Therefore, in markets with higher hospital competition, we find that, conditional on our set of controls, patients admitted to a VI hospital have 9.57 percentage point lower readmission risk relative to patients admitted to standard hospitals.

Table 6 shows that, after accounting for patient selection, vertically integrated hospitals lead to 10.9% shorter length of stay as compared to standard hospitals. Appendix ?? reveals an even stronger pattern under the stricter hospital definition. In unconcentrated markets, vertically integrated hospitals reduce length of stay by 13.9% relative to non-integrated hospitals, a magnitude that is slightly larger than the effect estimated in the full sample. These findings suggest that the observed reduction in readmission rates at vertically integrated hospitals is not driven by longer inpatient stays, but rather reflects more efficient care delivery.

Full Integration Results Table 7 reports the results of the baseline full integration specification in equation Equation 4. The specification includes both an indicator for full integration and an indicator for treatment at a VI hospital, so the coefficient on full integration

Table 5: Readmission: VI Hospital (Baseline Definition)

	Full Sample		Unconcentrated Markets		Concentrated Markets	
	OLS	IV	OLS	IV	OLS	IV
VI Hospital	0.0169*** (0.0052)	-0.0084 (0.0171)	0.0158 (0.0100)	-0.0957** (0.0376)	0.0152** (0.0062)	-0.0030 (0.0179)
Observations	177,648	177,648	38,168	38,168	138,861	138,861
KP first-stage F		138.9690		40.1930		118.7170
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Diagnosis FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Plan FE	Yes	Yes	Yes	Yes	Yes	Yes
Hospital FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Columns (1)–(2) use the full estimation sample; columns (3)–(4) restrict to unconcentrated markets with hospital HHI < 1500; columns (5)–(6) restrict to concentrated markets with hospital HHI $\geq$ 1500. Hospital, plan, county, diagnosis, and year fixed effects are included in all specifications. The IV specifications instrument hospital treatment with differential distance and ambulance-based destination preferences. Standard errors clustered at the hospital level are reported in parentheses. Stars denote $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

Table 6: Length of Stay: VI Hospital (Baseline Definition)

	Full Sample		Unconcentrated Markets		Concentrated Markets	
	OLS	IV	OLS	IV	OLS	IV
VI Hospital	-0.0179 (0.0113)	-0.1094*** (0.0390)	-0.0581** (0.0259)	-0.1390* (0.0741)	-0.0093 (0.0126)	-0.0952** (0.0437)
Observations	117,333	117,333	24,740	24,740	91,816	91,816
KP first-stage F		139.4000		40.2070		115.7970
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Diagnosis FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Plan FE	Yes	Yes	Yes	Yes	Yes	Yes
Hospital FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The outcome is $\log(LOS)$. Columns (1)–(2) use the full estimation sample; columns (3)–(4) restrict to unconcentrated markets with hospital HHI < 1500; columns (5)–(6) restrict to concentrated markets with hospital HHI $\geq$ 1500. Hospital, plan, county, diagnosis, and year fixed effects are included in all specifications. The IV specifications instrument hospital treatment with differential distance and ambulance-based destination preferences. Standard errors clustered at the hospital level are reported in parentheses. Stars denote $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

captures the incremental effect of sponsor-plan enrollment conditional on receiving care at a VI hospital.

The OLS estimates for both full integration and VI hospital are positive across samples, consistent with positive selection into integrated systems. In contrast, the IV estimates are highly imprecise and statistically insignificant for both coefficients in all samples. The first-stage statistics are weak, particularly in the unconcentrated markets, limiting the ability to draw causal conclusions.

Taken together, these results provide no evidence of an additional reduction in readmission rates from enrollment in a hospital-sponsored plan beyond the effect of being treated at a VI hospital. More broadly, we do not find robust evidence that either full integration or hospital-level integration reduces readmission rates in this specification.

Appendix [Table 8](#) leads to the same conclusion for length of stay. There is no evidence that fully integrated patients experience systematically different LOS once treatment at a VI hospital is already accounted for, and the first stage is again weakly identified. Taken together, the full-integration results are consistent with the idea that VI hospitals do not differentially manage patients based on whether they are enrolled in the sponsor’s plan. The lower readmission risk in VI hospitals appear to reflect hospital-wide practice changes rather than differential care by plan enrollment.

Overall, these estimates support four conclusions. First, in markets where hospital competition is stronger, hospital-plan integration significantly reduce readmission risk. Second, the reduction in readmission risk is not driven by longer inpatient stay at the hospital. Third, the lower readmission risk is driven hospital-wide care practice changes at the VI hospital. Finally, while we cannot rule out an effect through VI-plan enrollment, our results provide no evidence of an additional effect from enrollment in a VI-plan once we control for the admitting hospital.

6 Conclusion

This paper studies how hospital–plan integration affects 30-day unplanned readmissions risk, and aims at disentangling the effect of two treatment margins: the admission to VI-hospital and the enrollment in a VI-plan. Aggregate hospital-level data on Fee-for-Service beneficiaries indicate that VI-status adoption leads to hospital-wide care delivery that reduce readmission rate. Moreover, the event study analysis rules out that these results are driven by differential trends that precede VI-status adoption. We assume that the latter is also informative for pre-trend among Medicare Advantage beneficiaries. A limitation of aggregate FFS data is that we cannot control for endogenous patient selection across hospitals.

Table 7: Readmission: Full Integration (VI_full and VI_hosp)

	Full Sample		Unconcentrated Markets		Concentrated Markets	
	OLS	IV	OLS	IV	OLS	IV
Full integration	0.0228*** (0.0083)	-0.1093 (0.7457)	0.0275* (0.0154)	-1.5817 (1.9556)	0.0183* (0.0099)	-0.3927 (0.7791)
VI Hospital	0.0167*** (0.0052)	-0.0039 (0.0226)	0.0157 (0.0100)	-0.0288 (0.0765)	0.0151** (0.0062)	-0.0049 (0.0235)
Observations	177,648	79,076	38,168	14,303	138,861	64,293
KP first-stage F		2.529		0.407		1.989
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Diagnosis FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Plan FE	Yes	Yes	Yes	Yes	Yes	Yes
Hospital FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports estimates from specifications including both an indicator for full integration (VI_full) and an indicator for treatment at a vertically integrated hospital (VI_hosp). Columns (1)–(2) use the full estimation sample; columns (3)–(4) restrict to unconcentrated markets with hospital HHI < 1500; columns (5)–(6) restrict to concentrated markets with hospital HHI $\geq$ 1500. Hospital, plan, county, diagnosis, and year fixed effects are included in all specifications. The IV specifications instrument both VI_full and VI_hosp using interaction instruments and differential distance measures. Standard errors clustered at the hospital level are reported in parentheses. Stars denote $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

We use a pooled sample of two years of beneficiary-level Medicare Advantage data and implement an IV design to address beneficiaries selection across hospital and plans. Our set of instruments generate quasi-experimental variation in VI hospital assignment and VI-plan enrollment. We control for plan fixed effect and hospital fixed effects, and we leverage within hospital variation in the VI-status over time to identify the effect of admission to a VI-hospital. After accounting for patient selection, we find little evidence of an average hospital-side effect in the full sample. However, in unconcentrated hospital markets, admission to a vertically integrated hospital substantially reduces readmission risk relative to admission to a comparable non-integrated hospital. Moreover, we also show that this lower readmission risk is not driven by longer inpatient length of stay.

The plan-side identification is weaker since we cannot control for plan fixed effects which would absorb the VI-plan effect. While we cannot rule out plan side effects, our results shows that, after conditioning on the admitting hospital, there is no evidence that enrollment in a VI plan significantly changes readmission risk. We also find no evidence that beneficiaries who are both enrolled in a VI plan and treated at the sponsoring hospital experience an additional improvement beyond the effect of treatment at a VI hospital. However, those full-integration estimates are weakly identified and should be interpreted cautiously. Taken together, this evidence suggests that the benefits of hospital-plan integration arise primarily through hospital-wide changes in care delivery rather than through differential treatment based on plan enrollment.

These findings contribute to the debate on the efficiency gains from hospital-insurer integration. Integration can raise legitimate concerns about hospital market power, but our results here suggest that, in competitive hospital markets, it reduces significantly readmission risk without increasing inpatient stay. More broadly, the paper shows why it is important to separate hospital-side from plan-side effects when evaluating integrated payer-provider organizations. Because the cleanest hospital-side identification comes from high-acuity emergency admissions, the results speak most directly to that segment of inpatient care and need not generalize one-for-one to the full universe of hospital admissions. A natural next step is to examine whether similar effects arise in more discretionary episodes and for other outcomes, including post-acute spending, mortality, and total episode costs.

References

- Afendulis, C. C., Chernew, M. E., and Kessler, D. P. (2017). The effect of medicare advantage on hospital admissions and mortality. *American Journal of Health Economics*, 3(2):254–279.
- Agafiev Macambira, D., Geruso, M., Lollo, A., Ndumele, C. D., and Wallace, J. (2026). The private provision of public services: Evidence from random assignment in medicaid. *American Economic Review*, 116(6):2038–2084.
- Baicker, K., Chernew, M. E., and Robbins, J. A. (2013). The spillover effects of medicare managed care: Medicare advantage and hospital utilization. *Journal of Health Economics*, 32(6):1289–1300.
- Bejarano, G., Ryan, A., Trivedi, A., and Meyers, D. J. (2024a). Vertical integration and care experiences among medicare advantage beneficiaries. *JAMA Network Open*, 7(10):e2438994.
- Bejarano, G., Ryan, A., Trivedi, A., Offiaeli, K., and Meyers, D. J. (2024b). Medicare advantage plan and health system vertical integration, 2011-2020. *JAMA Network Open*, 7(7):e2423733.
- Berenson, R. A., Sunshine, J. H., Helms, D., and Lawton, E. (2015). Why medicare advantage plans pay hospitals traditional medicare prices. *Health Affairs*, 34(8):1289–1295.
- Callaway, B. and Sant’Anna, P. H. (2021). Difference-in-differences with multiple time periods. *Journal of econometrics*, 225(2):200–230.
- Chandra, A., Finkelstein, A., Sacarny, A., and Syverson, C. (2016). Health care exceptionalism? performance and allocation in the us health care sector. *American Economic Review*, 106(8):2110–2144.
- Chen, Y.-H., Kelleher, C. M., O’Donnell, W. J., Barron-Cervantes, N. M., and Chang, D. C. (2026). Growing trends in the payvider model and us hospital systems owning health plans. *JAMA*, 335(24):2162–2164.
- Chernew, M., DeCicca, P., and Town, R. (2008). Managed care and medical expenditures of medicare beneficiaries. *Journal of Health Economics*, 27(6):1451–1461.
- Cooper, Z., Doyle Jr, J. J., Graves, J. A., and Gruber, J. (2022). Do higher-priced hospitals deliver higher-quality care? Technical report, National Bureau of Economic Research.

- Cuesta, J. I., Noton, C. E., and Vatter, B. (2024). Vertical integration and plan design in healthcare markets. NBER Working Paper 32833, National Bureau of Economic Research. Revised May 2025.
- Curto, V., Einav, L., Finkelstein, A., Levin, J., and Bhattacharya, J. (2019). Health care spending and utilization in public and private medicare. *American Economic Journal: Applied Economics*, 11(2):302–332.
- Cutler, D. M., McClellan, M., and Newhouse, J. P. (2000). How does managed care do it? *The RAND Journal of Economics*, 31(3):526–548.
- Desai, N. R., Ross, J. S., Kwon, J. Y., Herrin, J., Dharmarajan, K., Bernheim, S. M., Krumholz, H. M., and Horwitz, L. I. (2016). Association between hospital penalty status under the hospital readmission reduction program and readmission rates for target and nontarget conditions. *JAMA*, 316(24):2647–2656.
- Doyle Jr, J. J., Graves, J. A., Gruber, J., and Kleiner, S. A. (2015). Measuring returns to hospital care: Evidence from ambulance referral patterns. *Journal of Political Economy*, 123(1):170–214.
- Frakt, A. B., Pizer, S. D., and Feldman, R. (2013). Plan–provider integration, premiums, and quality in the medicare advantage market. *Health Services Research*, 48(6):1996–2013.
- Gaynor, M., Ho, K., and Town, R. J. (2015). The industrial organization of health-care markets. *Journal of Economic Literature*, 53(2):235–284.
- Glied, S. and Zivin, J. G. (2002). How do doctors behave when some (but not all) of their patients are in managed care? *Journal of Health Economics*, 21(2):337–353.
- Gupta, A. (2021). Impacts of performance pay for hospitals: The readmissions reduction program. *American Economic Review*, 111(4):1241–1283.
- Kessler, D. P. and McClellan, M. B. (2000). Is hospital competition socially wasteful? *The Quarterly Journal of Economics*, 115(2):577–615.
- Lin, H., McCarthy, I. M., and Richards, M. (2021). Hospital pricing following integration with physician practices. *Journal of Health Economics*, 77:102444.
- Ma, Y., Zheng, J., Orav, E. J., Figueroa, J. F., Meyers, D. J., and Tsai, T. C. (2026). Hospital-operated medicare advantage plans and care utilization and outcomes after inpatient admissions. *JAMA Network Open*, 9(8):e2632079.

- Maeda, J. L. K. and Nelson, L. (2018). How do the hospital prices paid by medicare advantage plans and commercial plans compare with medicare fee-for-service prices? *Inquiry*, 55:1–8.
- Meyers, D. J., Mor, V., and Rahman, M. (2020). Provider integrated medicare advantage plans are associated with differences in patterns of inpatient care: An assessment of if integrated medicare advantage plans lead to differences in the processes that take place during hospitalizations and to differences in patient outcomes. *Health Affairs*, 39(5):843–851.
- Ody, C., Msall, L., Dafny, L. S., Grabowski, D. C., and Cutler, D. M. (2019). Decreases in readmissions credited to medicare’s program to reduce hospital readmissions have been overstated. *Health Affairs*, 38(1):36–43.
- Park, S., Langellier, B. A., and Meyers, D. J. (2023). Differences between integrated and non-integrated plans in medicare advantage. *Health services research*, 58(3):560–568.
- Weiss, A. J. and Jiang, H. J. (2021). Overview of clinical conditions with frequent and costly hospital readmissions by payer, 2018. Technical Report 278, Agency for Healthcare Research and Quality.
- Ziedan, E. and Kaestner, R. (2021). Did the hospital readmissions reduction program reduce readmissions? an assessment of prior evidence and new estimates. *Evaluation Review*, 45(6):359–411.
- Zuckerman, R. B., Sheingold, S. H., Orav, E. J., Ruhter, J., and Epstein, A. M. (2016). Readmissions, observation, and the hospital readmissions reduction program. *New England Journal of Medicine*, 374(16):1543–1551.

A Event Study Estimates and Pre-Trend Tests

This appendix reports event-study estimates corresponding to the baseline difference-in-differences specification using the Callaway and Sant’Anna (2021) estimator. The goal of these results is to assess the validity of the identifying assumptions, in particular the absence of differential pre-treatment trends, and to illustrate the dynamic evolution of outcomes following vertical integration (VI) adoption.

The main text showed pre-trends for the unconcentrated market. Figures 3 and 4 present analogous event-study estimates for the full sample and for concentrated markets, respectively. In both cases, the pre-treatment coefficients are generally small and fluctuate around zero, with no clear monotonic trend, suggesting limited evidence of systematic differences in outcomes prior to VI adoption. We do not find evidence for VI hospitals reducing readmission rates in concentrated market or the full sample.

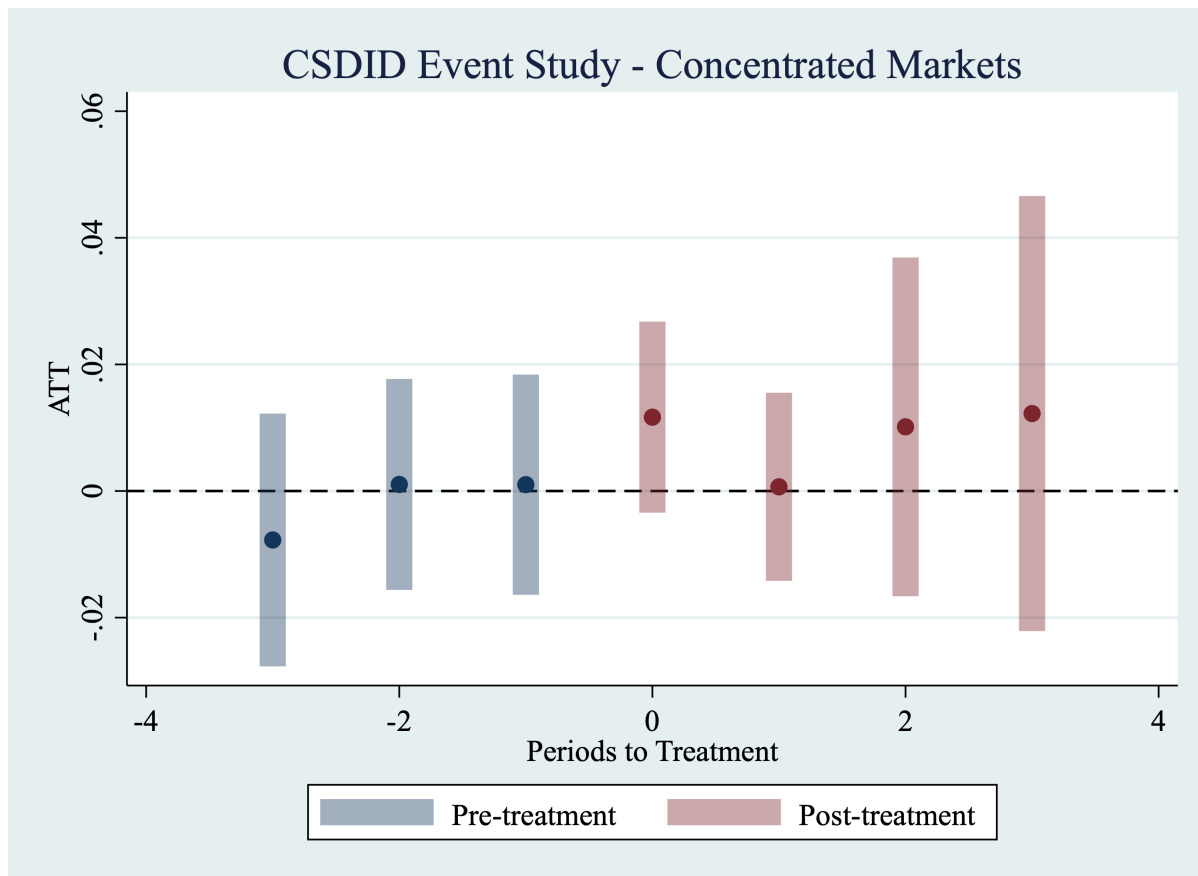


Figure 3: FFS event study around first hospital VI adoption

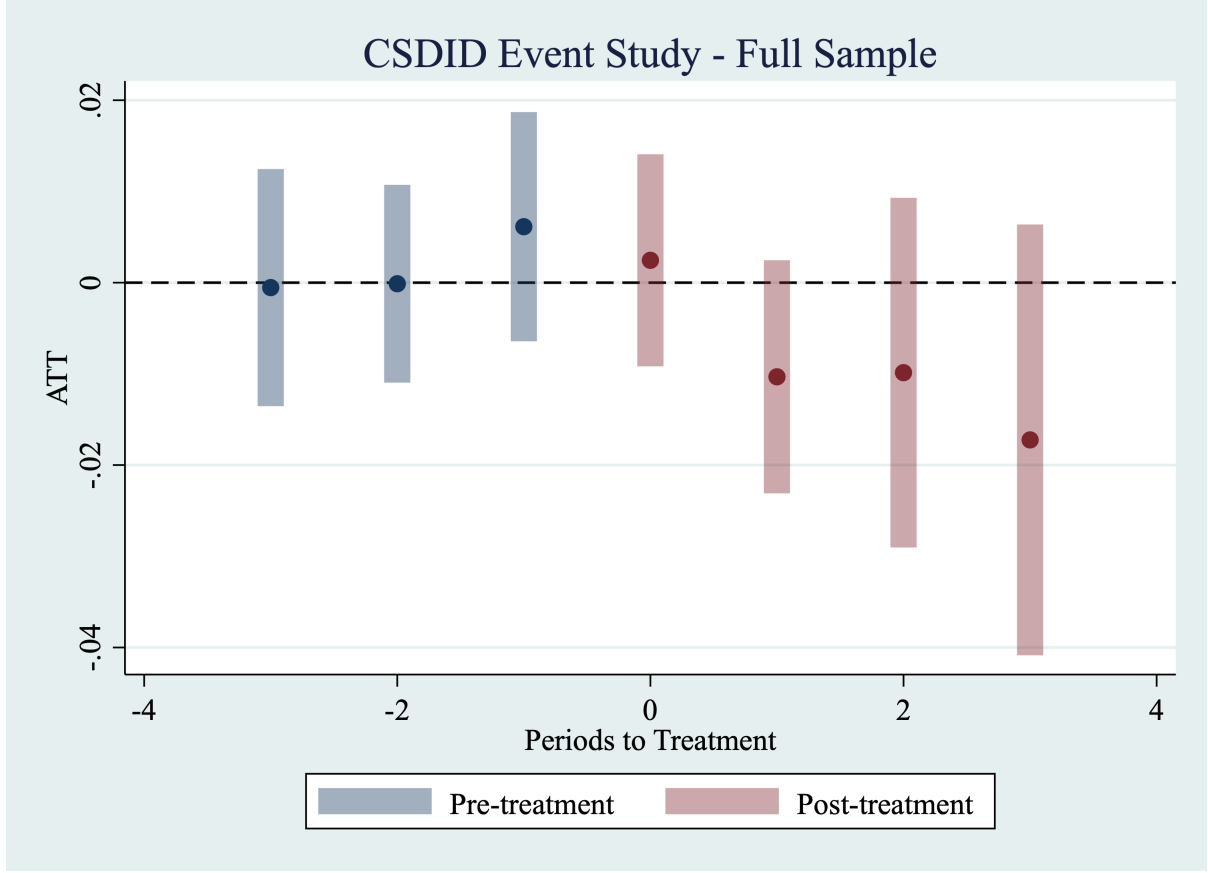


Figure 4: FFS event study around first hospital VI adoption

B Additional Results Tables

This appendix collects the additional tables. The first group reports validation exercises for the hospital-side margin using a stricter VI-hospital definition based on sponsor-plan enrollment relative to hospital beds. The second group reports length-of-stay results for the plan-side and full-integration margins. These tables support the interpretation in the main text: the hospital-side readmission effect is strongest in unconcentrated markets and is not driven by longer stays, whereas the evidence for additional plan-side or full-integration effects is substantially weaker.

Table 8: Length of Stay: Full Integration

	Full Sample		Unconcentrated Markets		Concentrated Markets	
	OLS	IV	OLS	IV	OLS	IV
Full integration	-0.0137 (0.0171)	1.6952 (1.6295)	-0.0311 (0.0352)	1.5148 (1.5373)	0.0047 (0.0184)	1.5148 (1.5373)
VI Hospital	-0.0264** (0.0107)	-0.1378*** (0.0508)	-0.0789*** (0.0239)	-0.1104** (0.0541)	-0.0120 (0.0123)	-0.1104** (0.0541)
Observations	177,648	79,076	38,168	—	138,861	64,293
KP first-stage F		2.529		—		1.989
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Diagnosis FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Plan FE	Yes	Yes	Yes	Yes	Yes	Yes
Hospital FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The outcome is $\log(LOS)$. The table reports estimates from specifications including both an indicator for full integration (VI_full) and an indicator for treatment at a vertically integrated hospital (VI_hosp). Columns (1)–(2) use the full estimation sample; columns (3)–(4) restrict to unconcentrated markets with hospital HHI < 1500; columns (5)–(6) restrict to concentrated markets with hospital HHI $\geq$ 1500. Hospital, plan, county, diagnosis, and year fixed effects are included in all specifications. The IV specifications instrument both VI_full and VI_hosp using interaction instruments and differential distance measures. Standard errors clustered at the hospital level are reported in parentheses. Stars denote $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.